TASK 05-50-03-200-802-A
EFFECTIVITY: ALL

1. Off - Temperature Envelope Flight - Inspection/Check

A. General

(1) This task contains the necessary actions for aircraft operated outside the flight envelope temperature given in the AFM.

(2) This section gives the necessary actions after a flight envelope temperature exceedance is reported by crew, or detected by flight data analysis.

(3) The flight envelope temperature exceedance is presented in the AFM - Section 2 Limitations, and is reproduced in the Temperature Allowance Chart Figure 615, together with the thresholds and exceedance zones specified on this procedure.

(4) If the temperature given in Figure 615 is exceeded during flight, do the steps at the Job Set-up SUBTASK 160-002-A.

(5) If the temperature given in Figure 615 be respected during flight, no further action is necessary, go to SUBTASK 000-002-A, any other condition

(6) The operator can do the off-temperature envelope flight analysis, but if you choose to send the data for EMBRAER analysis, it is recommended that you send the TCRF file.

(7) This maintenance technician is necessary to do this task:

- 1 - Electrical/Avionics Systems

B. References

REFERENCE	*DESIGNATION*
AMM TASK 05-50-50-200-818-A/600	Engine After Flying Outside of Total Air Temperature (TAT) Limits

C. Job Set-Up

SUBTASK 160-002-A

(1) If the temperature given in Figure 615 is exceeded in all cold condition areas during the flight, do the steps that follow:

NOTE: The operator can do the analysis of the flight operation data, but if you choose to send the data for an EMBRAER analysis, it is recommended that you send the TCRF file

(a) Do the skin temperature calculation.SUBTASK 160-004-A

NOTE: The temperature limit for skin is -54 °C.

(b) Do the engine inspection.SUBTASK 200-002-A

D. Total Air Temperature (TAT) calculation for temperature envelope exceedance check

SUBTASK 160-004-A

(1) NOTE: The operator can do the analysis of the flight operation data, but if you choose to send the data for an EMBRAER analysis, it is recommended that you send the TCRF file.

Determine the coldest Total Air Temperature (TAT) experienced during the cold condition exceedance in Figure 615.

(a) Total Air Temperature (TAT) can be obtained through TCRF file (total AirTemperature-1), or;

(b) Do the Total Air Temperature (TAT) calculation, as follow:

- TAT = SAT [1+(0.2 M^2)], where;
  NOTE: The temperature must be converted to K (Kelvin) for calculation, and then changed to °C (Celsius) for verification.
- TAT = Total Air Temperature [K]
- SAT = Static Air temperature [K]
- M = Mach Number

E. Skin temperature check/inspection

SUBTASK 160-003-A

(1) Do the skin temperature calculation SUBTASK 160-004-A.

(2) Is the calculated skin temperature -54 °C or colder?

(a) YES - Contact the RTS and supply TCRF data for analysis and recommendations.

(b) NO - No more actions for skin are necessary.

F. Engine temperature check/inspection

SUBTASK 200-002-A

(1) Do the engine temperature calculation SUBTASK 160-004-A.

(2) Was the total air temperature colder than -54°C?

(a) YES - Do the engine inspection (Refer to AMM TASK 05-50-50-200-818-A/600).

NOTE: If the TAT is colder than -63°C, contact Pratt and Whitney.

(b) NO – No more actions are necessary

G. Job Close-Up

SUBTASK 000-002-A

(1) Put the aircraft back to its initial condition.

EFFECTIVITY: ALL
Temperature Allowance Chart
Figure 615